

A Novel Multihead Attention-Enhanced Convolutional Neural Network-Long Short-Term Memory Encoder-Decoder for Oil Production Forecasting

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Summary

Oil production forecasting is crucial in reservoir engineering for decision support and strategy optimization. Traditional methods struggle with capturing complex temporal dependencies and nonlinear relationships in production data, especially when other influencing factors are incomplete, missing, or contain unreliable outliers. Even after data preprocessing, these features often remain inadequate. Additionally, most previous researches have focused on single-step forecasting, which is insufficient for comprehensive planning and decision-making. To address these challenges, this paper proposes a novel hybrid encoder-decoder deep learning framework, named MA-CLSTM-ED, which combines convolutional neural network (CNN), long short-term memory (LSTM) network, and multihead attention mechanism to reveal intricate patterns and insights within the oil production data. This model is specifically designed for scenarios where only production data are available, using innovative feature engineering methods that extract first-order differences and cumulative sum data to enhance prediction accuracy. The Rime optimization algorithm (RIME), inspired by the physical phenomenon of rime-ice formation, is used for hyperparameter optimization, further improving the model's efficiency and performance. Experimental results demonstrate that the proposed MA-CLSTM-ED model significantly outperforms traditional methods in multistep oil production forecasting, maintaining high prediction accuracy even in challenging data conditions. The integration of differential and cumulative data as input features reduces the impact of minor parameter changes on predictions, decreasing forecast chaos and improving stability and accuracy. Multiple experiments validate the effectiveness and applicability of the proposed methods, highlighting the model's potential in providing more accurate and reliable tools for oil production forecasting.

Introduction

Oil, often referred to as the “blood of industry,” serves as a primary energy source and a crucial chemical raw material. Its production is vital for effective reservoir management and optimization. Accurate oil production forecasting enhances efficiency and economic benefits by optimizing strategies and plans, and it helps mitigate potential risks, ensuring sustainable development of oil fields.

Traditional methods, such as decline curve analysis and reservoir simulation, are commonly used for forecasting. However, decline curve analysis struggles with dynamic factors and requires stable production conditions (Jaber et al. 2019), while reservoir simulation demands significant computational resources and time (Liu et al. 2024). Both methods face challenges, such as sparse data, complex nonlinear relationships, and dynamic system changes. Given that interventions and measures in petroleum engineering are inherently reflected in the production data, this study proposes a data-driven machine learning approach. When the data set is sufficiently large and comprehensive, the various patterns of abrupt changes and gradual shifts can be effectively captured and modeled by learning from historical production data.

With advancements in machine learning and deep learning, methods such as artificial neural networks (Aminian and Ameri 2005; Li et al. 2013; Muradkhanli 2018; Negash and Yaw 2020), support vector machines (Wang and Miao 2012; Ccoicca 2013; Khan et al. 2019), and random forests (Han et al. 2020; Gonçalves et al. 2022; Qiao et al. 2023) have achieved significant success in oil production forecasting. Despite advancements, oil production forecasting, as a time-series problem, poses significant challenges for traditional machine learning methods due to temporal dependencies. Recurrent neural networks (RNNs), introduced by Elman (1990), were developed to handle these challenges.

However, RNNs suffer from the vanishing gradient problem, which the LSTM model (Hochreiter and Schmidhuber 1997) addresses by capturing long-term dependencies more effectively. Still, single models such as LSTM often struggle with the complexity and nonlinearity of oil production data. For instance, LSTM models were used for shale gas production forecasting by both Lee et al. (2019) and Yang et al. (2022). In Lee's study, monthly production and shutdown features were incorporated, with L2 regularization and dropout techniques utilized to prevent overfitting and enhance prediction accuracy. Similarly, Yang's approach combined production data with traditional methods, leading to significant improvements in accuracy and robustness. Sagheer and Kotb (2019) introduced a deep LSTM architecture, which significantly enhanced the accuracy and robustness of oil production time series forecasting, particularly excelling in handling nonlinear and long interval data. These studies underscore the utility of LSTM in oil production forecasting while also highlighting inherent limitations when relying solely on LSTM.

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To address these limitations, researchers have turned to hybrid models. Studies indicate that single models have limited predictive performance for complex time series data, while hybrid models incorporating deep learning methods, such as deep LSTM, ensemble empirical mode decomposition combined with LSTM, and graph convolutional network combined with LSTM, demonstrate higher predictive accuracy and robustness. These hybrid models effectively extract complex patterns and long-term dependencies, providing more accurate and reliable tools for oil production forecasting.

However, these existing hybrid models still face difficulties in capturing the full complexity of oil production data, particularly in long-term forecasting scenarios. Tadjer et al. (2021) compared DeepAR, a deep learning-based autoregressive model (Salinas et al. 2020) and Prophet (Taylor and Letham 2018) for short-term forecasting, showing DeepAR's superiority in long-term pattern memory and prediction interval coverage by combining LSTM and gated recurrent unit for joint learning. Despite these advances, further improvements are needed to fully address the inherent nonlinearity and long-term dependencies present in oil production data. Liu et al. (2020) proposed the ensemble empirical mode decomposition LSTM method, which decomposes oil production series into intrinsic mode functions, enhancing LSTM's ability to capture pattern changes. Experiments showed that the ensemble empirical mode decomposition LSTM method effectively handles nonlinear and high-noise data. Du et al. (2022) proposed the graph convolutional network-LSTM model, integrating graph convolutional network for spatial patterns and LSTM for temporal correlations, which outperformed single models in handling well interference and complex data.

Recent studies highlight the effectiveness of combining CNNs, LSTM, and attention mechanisms in oil production forecasting. The particle swarm optimization-CNN-LSTM model proposed by Li et al. (2022), optimized using particle swarm optimization, demonstrated higher accuracy and robustness. Similarly, Song et al. (2023) developed a CNN-LSTM hybrid model for tight reservoir forecasting that outperformed backpropagation neural network, radial basis function network, RNN, and standalone LSTM models across multiple data sets. Additionally, Kumar et al. (2023) proposed an attention-LSTM hybrid model for oil production forecasting. The attention mechanism highlights important information by assigning weights to input sequences and using a dynamic floating window technique to prevent data leakage, further improving model adaptability. This model showed significant advantages over traditional models in oil and gas forecasting. These examples illustrate the necessity of hybrid models for accurate oil production forecasting. By leveraging the strengths of each component, hybrid models offer improved prediction accuracy and robustness, addressing the limitations of single models.

Nevertheless, data-driven models still face significant challenges. Oil production is influenced by multiple factors, with nonlinear relationships making accurate modeling difficult. Minor parameter variations can significantly impact long-term forecasts, causing chaotic predictions (Deng et al. 2024; Fan et al. 2024). Extracting time series information is challenging due to the complex interrelationships over time. To overcome these challenges and enhance model performance, we innovatively extracted first-order cumulative sum and first-order difference as input features. The first-order cumulative sum helps the model better capture long-term trends, while the first-order difference is effective in capturing short-term fluctuations.

While much of this research focuses on short-term predictions, oil companies require long-term planning and decision-making. Single-step predictions offer limited practicality. Companies need to forecast production changes over extended periods to allocate resources and adjust strategies comprehensively. Ning et al. (2022) compared autoregressive integrated moving average, LSTM, and Prophet models in multistep forecasting under different conditions, highlighting each model's advantages but focusing on trend research. Werneck et al. (2022) applied deep learning methods, such as gated recurrent unit 2, gated recurrent unit convolutional sequence to sequence, and CNN, to multistep forecasting, combining historical sensor data from production and injection wells to predict multiple future points. No single network excelled in all scenarios, indicating the need for specific approaches in oil production forecasting. Du et al. (2024) discussed multistep forecasting of oil production, focusing on annual US production. However, the study did not delve into detailed monthly production data for individual wells. Since production planning requires more granular forecasts at the well level over shorter time frames, such macrolevel predictions offer limited practical utility for operational guidance.

Given the challenges in accurately forecasting oil production, particularly in scenarios where only production data are available and existing methods struggle with long-term forecasting and complex pattern recognition, this paper proposes a novel hybrid encoder-decoder deep learning framework, named MA-CLSTM-ED, which combines the CNN, LSTM network, and the multihead attention mechanism. The encoder-decoder architecture is particularly suited for sequential data prediction tasks, allowing the model to effectively capture temporal dependencies and patterns over multiple steps (Alzubaidi et al. 2021).

To improve prediction accuracy, we introduce an innovative feature engineering approach that incorporates first-order differences and cumulative sum data, enabling the model to better capture both short-term fluctuations and long-term trends. Furthermore, the RIME is utilized for hyperparameter optimization, enhancing the model's efficiency and predictive performance.

The MA-CLSTM-ED model can incorporate additional covariates, such as geological and operational data. However, this study intentionally focuses on historical production values. This choice is motivated by the inconsistent availability and quality of covariate data across different wells. By relying solely on historical data, the model achieves simplicity and broad applicability across diverse production environments.

Experimental results demonstrate that the MA-CLSTM-ED model significantly outperforms traditional methods in multistep oil production forecasting, maintaining high accuracy even under challenging data conditions. By integrating differential and cumulative data as input features, the model mitigates the impact of minor parameter variations, reducing forecast instability and enhancing prediction robustness. These findings underscore the effectiveness of our proposed feature extraction techniques and optimization strategies, highlighting the model's potential as a reliable tool for oil production forecasting. While covariates, such as geological and operational data, are indeed crucial for enhancing forecast accuracy, future work will focus on incorporating these covariates to further improve the model's performance.

In summary, this study introduces a RIME-optimized MA-CLSTM-ED model that leverages the strengths of CNN, LSTM, and multihead attention for enhanced multistep forecasting of single-well oil production. By incorporating differential and cumulative data as input features, the model effectively captures complex patterns and long-term dependencies, resulting in more stable and accurate long-term predictions.

The rest of this paper is organized as follows: The "Methodology" section provides a detailed explanation of the proposed model. The "Experiments" section covers the data set description, preprocessing steps, and the evaluation metrics used. The "Experimental Results and Analysis" section presents and discusses the findings from the experiments. Finally, the "Concluding Remarks" section summarizes the contributions of this paper and discusses future research directions.

Methods

Problem Description. Oil production data are inherently time series data, which consist of measurements taken at successive points in time. The time element specifies the period or point in time, while the value element can correspond to one or multiple variables. A single variable forms a univariate time series, whereas multiple variables form a multivariate time series.

In this study, the multivariate time series of oilwell production and its related inputs is expressed as $Y = [y_1, y_2, \dots, y_T] \in R^{N \times T}$, where T is the number of timesteps, and N is the number of variables. Since the input data for the model established in this paper only consider historical production data and the first-order difference and cumulative sum data calculated from historical production data, Y can be simplified to $Y = [y_1, y_2, \dots, y_T] \in R^{3 \times T}$. In this expression, $y_t \in R^3$ represents the slice of Y at timestep t . Given a historical observation window of length T , $Y^{\text{in}} = [y_1^{\text{in}}, y_2^{\text{in}}, \dots, y_T^{\text{in}}]$, and a forecast time span τ , the task is to predict the values for the next τ steps $Y^{\text{out}} = [y_1^{\text{out}}, y_2^{\text{out}}, \dots, y_\tau^{\text{out}}]$.

Feature Extraction Method. To enhance the performance of the oil production forecasting model, differential and cumulative sum data are introduced as features. These features help capture short-term fluctuations and long-term trends in the time series, thereby improving the accuracy and stability of time series forecasting. Extracting the first-order differential data of oil production removes trends and seasonal components, making the data more stationary. Extracting the first-order cumulative sum data reveals long-term trends and cumulative effects, helping the model capture long-term structures to predict long-term changes. The calculations for the first-order differential and first-order cumulative sums of oil production are shown in Eq. 1 and Eq. 2:

$$Di = y_i - y_{i-1}. \quad (1)$$

$$S = y_i + y_{i-1}. \quad (2)$$

Given the oil production data $Y_{t-k:t} = [y_{t-k}, y_{t-k-1}, \dots, y_t]$, the first-order differential of oil production is $Di_{t-k+1:t} = [y_{t-k-1} - y_{t-k}, \dots, y_t - y_{t-1}]$, and the first-order cumulative sum is $S_{t-k+1:t} = [y_{t-k-1} + y_{t-k}, \dots, y_t + y_{t-1}]$.

Machine Learning Methods. CNN. CNNs are feedforward multilayer neural networks capable of processing multichannel parallel time series data. A CNN consists of filtering stages and a classifier (Widiputra et al. 2021). The filtering stages, including convolution and pooling operations, extract features from the input data. The classification stage comprises multiple fully connected layers, achieving thorough feature extraction through stacked convolution and pooling layers. **Fig. 1** illustrates the CNN network structure: Data enter the convolutional layer through the input layer, where convolution operations extract important features; then, they proceed to the pooling layer, reducing the number of parameters; finally, the extracted features are integrated through fully connected layers, outputting the prediction results.

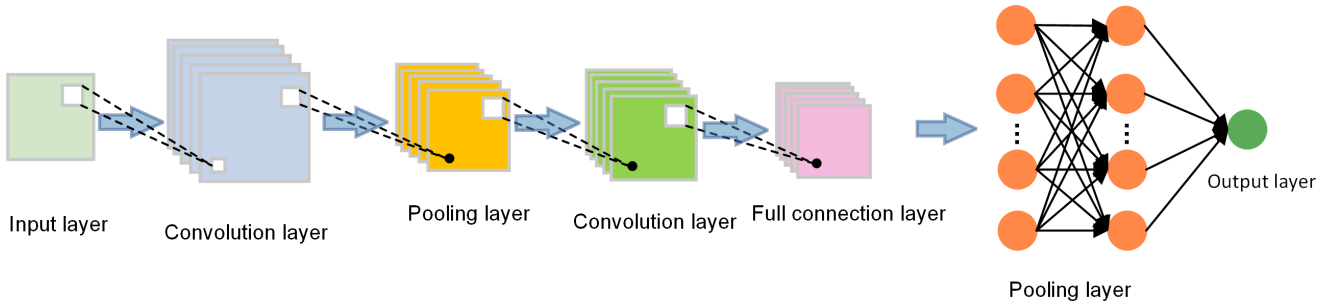


Fig. 1—Structure of CNN diagram.

CNNs can classify different types of input data and quickly extract effective features, capturing the potential relationships among oilwell production data, oilwell production differential data, and oilwell production cumulative data. This extraction of local features significantly enhances data processing capabilities. The operation of the convolutional layer is shown in Eq. 3:

$$M_j^N = f \left(\sum_i M_i^{N-1} * C_j^N + b_j^N \right), \quad (3)$$

where M_j^N is the j th feature map in the N th convolutional layer, $f(x)$ is the activation function, C_j^N is the j th convolution kernel in the N th convolutional layer, and b_j^N is the bias. CNNs excel at capturing local features but are weak in capturing long-distance temporal dependencies. To overcome this limitation and better capture the long-term dependencies in time series data, LSTM networks are introduced after CNN.

LSTM. RNNs consist of repeating neural network modules but are limited by issues such as short-term memory, gradient explosion, gradient vanishing, and long-term dependencies. LSTM networks, a variant of RNNs, overcome the gradient vanishing problem and handle long-term dependencies. The main structure of LSTM includes input gates, output gates, and forget gates, as shown in **Fig. 2** (Wang et al. 2022).

The forget gate processes the previous cell state c_{t-1} and the current input information x_t using an activation function σ to determine the extent of retention. The output of the forget gate f_t is shown in Eq. 4:

$$f_t = \sigma (X_t U_f + h_{t-1} W_f + b_f). \quad (4)$$

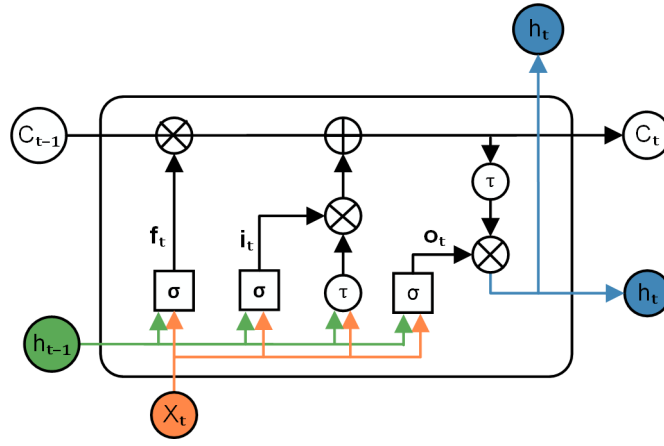


Fig. 2—LSTM basic architecture with forget gate (f_t), input gate (i_t), and output gate (o_t).

The output obtained from Eq. 4 varies between 0 and 1, where 0 means none of the learned information is retained, and 1 means all of it is retained. W_f and U_f are the weight matrices of the forget gate, b_f is the bias value of the forget gate, and σ is the sigmoid activation function. The input gate calculates the extent to which the current input information x_t is retained in the cell state c_t at the current time-step, as shown in Eq. 5 and Eq. 6:

$$i_t = \sigma(X_t U_i + h_{t-1} W_i + b_i). \quad (5)$$

$$\tilde{c}_t = \tanh(X_t U_c + h_{t-1} W_c + b_c). \quad (6)$$

Here, W_i and U_i represent the weight matrices of the input gate, $\tanh$ is the activation function, U_c and W_c are the weight matrices of the new memory cell, and b_i and b_c are the bias vectors of the input gate and the new memory cell, respectively. The cell state variable c_t at the current timestep consists of two parts, as shown in Eq. 7:

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t. \quad (7)$$

The output gate calculates the extent of the output of the cell state variable at the current timestep, as shown in Eq. 8 and Eq. 9:

$$o_t = \sigma(X_t U_o + h_{t-1} W_o + b_o). \quad (8)$$

$$h_t = o_t * \tanh(c_t). \quad (9)$$

The three gate structures perform matrix multiplication and nonlinear summation operations within the memory unit to ensure that the memory does not decay during the iteration process.

Encoder-Decoder Model Based on Attention Mechanism. This paper focuses on multistep forecasting of oil production, which falls under the category of seq2seq problems. Seq2seq, short for sequence to sequence, is a neural network architecture designed to process sequential data. It has been effectively used in various applications, including time series forecasting, where both the input and output are sequences. Encoder-decoder models have achieved good results in time series forecasting. As shown in **Fig. 3a**, a traditional encoder-decoder model first converts the input time series into a fixed-length vector through the encoder. The decoder then converts this vector back into the output sequence. The encoder and decoder can use various models such as LSTM, RNN, or CNN. While encoder-decoder models outperform single models like LSTM or RNN, they have limitations. The fixed-length vector connecting the encoder and decoder can cause early input information to be overshadowed by later information, especially for long sequences. Additionally, this vector may not fully capture the entire sequence's information. To overcome these limitations, an attention mechanism can be introduced into the encoder-decoder model (Wen and Li 2023), as shown in **Fig. 3b**. The attention mechanism allows the model to focus on relevant parts of the input sequence when generating each part of the output sequence, improving the representation of long sequences.

Overall Model Architecture. This paper proposes the MA-CLSTM-ED model for oil production forecasting, integrating CNN, LSTM, and an attention-based encoder-decoder model to leverage their respective strengths. The structure of this model is shown in **Fig. 4**.

The MA-CLSTM-ED model comprises the following layers:

Input layer: Accepts the preprocessed time series data $Y^{\text{in}} = [y_1^{\text{in}}, y_2^{\text{in}}, \dots, y_T^{\text{in}}]$, where T is the length of the input time series.

Encoder layer: Uses a CNN-LSTM encoder to extract correlations and local features from the production data, differential data, and cumulative data, while capturing long-term dependencies.

Attention layer: Calculates attention weights to focus on relevant parts of the input sequence.

Decoder layer: Decodes the weighted encoded data to generate the output sequence.

Output layer: Computes the predicted sequence for the next τ timesteps $Y^{\text{out}} = [y_1^{\text{out}}, y_2^{\text{out}}, \dots, y_\tau^{\text{out}}]$.

The encoder layer, attention layer, and decoder layer are described in detail below.

Encoder Layer. In the MA-CLSTM-ED model, the input vector $Y^{\text{in}} = [y_1^{\text{in}}, y_2^{\text{in}}, \dots, y_T^{\text{in}}]$ after preprocessing by the input layer is fed into the encoder layer for encoding. The encoder is composed of CNN and LSTM models. The calculations in the encoder layer are as follows:

CNN: First, the input vector is processed by the CNN to extract local features from the input sequence. For each timestep i , assuming the kernel size is K , the convolution operation is shown in Eq. 10:

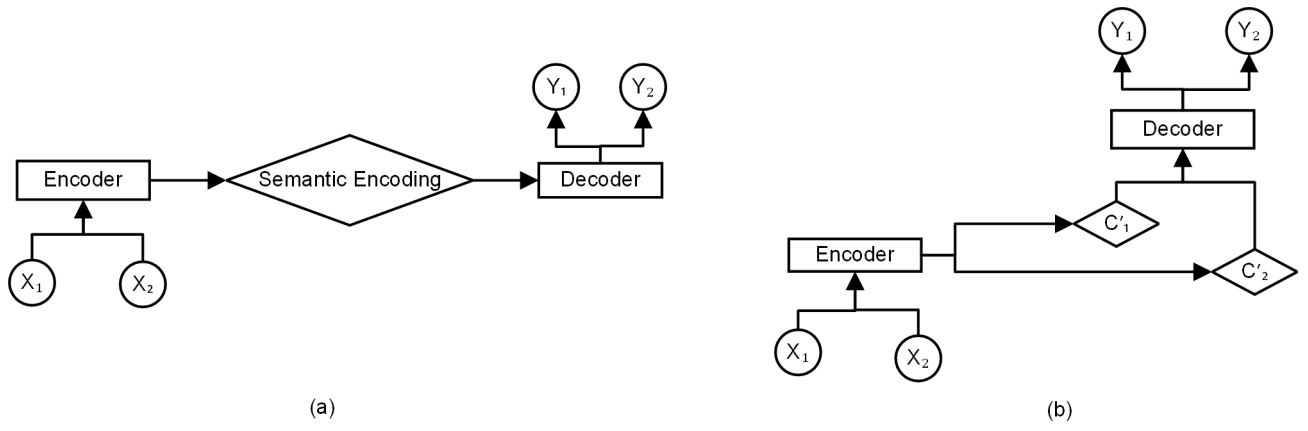


Fig. 3—Encoder-decoder models for time series forecasting: (a) encoder-decoder model and (b) encoder-attention-decoder model.

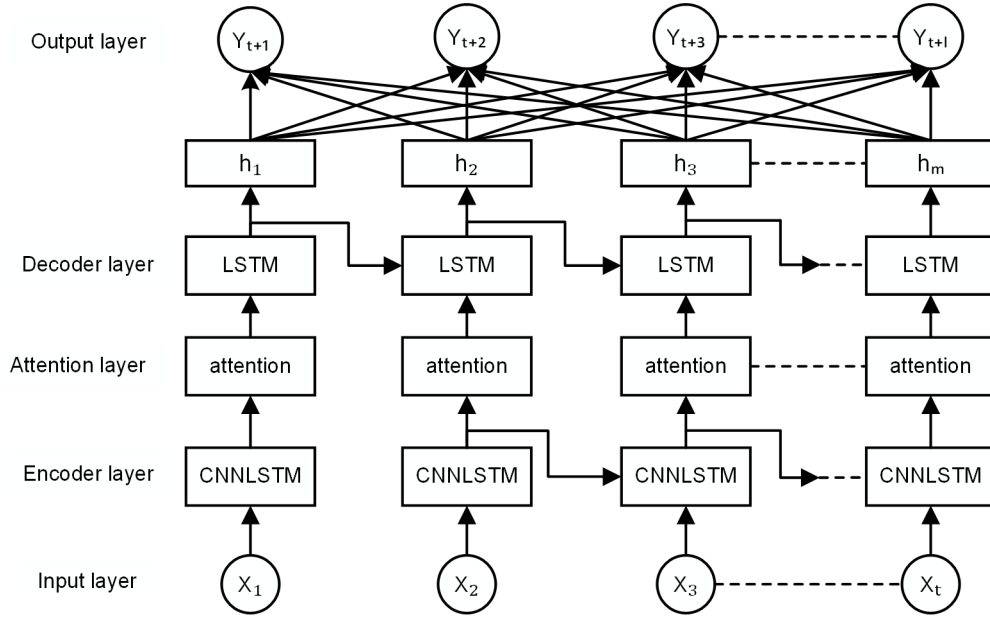


Fig. 4—Structure of CNNLSTM-attention-LSTM model.

$$C_i = f_{\text{cnn}}(Y_i^{\text{in}}), \quad (10)$$

where Y_i is the output of the input layer at time i , $1 \leq i \leq t$, C_i is the output of the convolutional layer at time i , f_{cnn} is the calculation function of the CNN model, and t represents the timesteps.

LSTM: The features extracted by the convolution operation are then input into the LSTM layer to capture long-term dependencies in the sequence. The calculation process of LSTM is shown in Eq. 11:

$$l_i = f_{\text{lstm}}(C_i, l_{i-1}), \quad (11)$$

where l_i represents the hidden state of the encoder layer at time i , $1 \leq i \leq t$, l_{i-1} is the hidden state at time $i-1$, t represents the timesteps, and f_{lstm} represents the overall calculation function of the input, forget, and output gates in the LSTM, as described in section “LSTM.”

Attention Layer. The vector output from the encoder layer is fed into the attention layer to calculate the attention distribution probability for each time series. Since the model in this paper uses LSTM as the decoder, when predicting the value y_i^{out} at time i , we know the hidden layer output value l_i obtained by the decoder before y_i^{out} . Therefore, the function $F(l_i, l_j)$ can be used to calculate l_i (i.e., the hidden state of the decoder at time i) and l_j (i.e., the hidden state of the encoder at time j). The attention weight calculation formulas are shown in Eq. 12 and Eq. 13:

$$e_{ij} = a(l_{i-1}, l_j), \quad (12)$$

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})} = \text{softmax}(e_{ij}), \quad (13)$$

where α_{ij} represents the attention weight, and e_{ij} represents the attention relevance value from time j to time i . The nonlinear function is used to compare the hidden state h_j at time j in the encoder with the hidden state l_{i-1} of the output sequence at time $i - 1$. The higher the match degree, the larger e_{ij} and α_{ij} , which means that more output at time i can be allocated to the input at time j , and the input at time j is more influenced by it.

Given an input time series $Y^{\text{in}}(y_1^{\text{in}}, y_2^{\text{in}}, y_3^{\text{in}})$, the attention allocation model assigns attention weights to the prediction time point sequence y_3^{in} as $a_{3j} = \{(y_1^{\text{in}}, 0.3), (y_2^{\text{in}}, 0.3), (y_3^{\text{in}}, 0.4)\}$. Similarly, each time point sequence in the target sequence also learns the corresponding attention distribution probability information of the original sequence. Thus, when generating each time point sequence y_i^{out} , the previously fixed semantic vector M is replaced by M'_i , which changes dynamically according to the current generated sequence. Therefore, the information of the time point sequence corresponding to $a_{3j} = \{(y_1^{\text{in}}, 0.3), (y_2^{\text{in}}, 0.3), (y_3^{\text{in}}, 0.4)\}$ is shown in Eq. 14:

$$M'_i = g(0.3 \times l_1, 0.3 \times l_2, 0.4 \times l_3), \quad (14)$$

where l_1 , l_2 , and l_3 are the output values of the encoder layer at each time point, and the function g represents the transformation function that integrates each time point sequence into a single timestep sequence, as shown in Eq. 15:

$$M'_i = \sum_{j=1}^t \alpha_{ij} h_j. \quad (15)$$

In essence, the function g is a weighted summation function. M'_i is the output vector of the attention layer at time i , h_j is the hidden state at time j , α_{ij} represents the attention weight from time j to time i , and t is the timestep.

Decoder Layer. The M'_i obtained from the attention mechanism layer is used as the input sequence for the decoder layer. It is processed to compute and output y_i , as shown in Eqs. 16 through 21:

$$f_i = \delta(W_f[y_{i-1}^{\text{out}}, l_i, M'_i] + b_f), \quad (16)$$

$$l_i = \delta(W_l[y_{i-1}^{\text{out}}, l_i, M'_i] + b_l), \quad (17)$$

$$Z_i = \tanh(W_c[y_{i-1}^{\text{out}}, l_i, M'_i] + b_c), \quad (18)$$

$$M_i = f_i \times M_{i-1} + l_i \times Z_i, \quad (19)$$

$$O_i = \delta(W_o[y_{i-1}^{\text{out}}, l_i, M'_i] + b_o), \quad (20)$$

$$y_i^{\text{out}} = O_i \times \tanh(M_i), \quad (21)$$

where f_i , l_i , and O_i represent the forget gate, input gate, and output gate, respectively. Z_i and M_i represent the cell state updates, M_{i-1} is the cell state at the previous timestep $i - 1$, and δ is the sigmoid activation function. W_f , W_l , W_c , and W_o are the input weights, while b_f , b_l , b_c , and b_o are the input biases. y_{i-1}^{out} is the predicted output of the decoder at time $i - 1$, l_i is the hidden state of the decoder at time i , and M'_i is the output vector of the attention layer at time i . The predicted output y_i^{out} , after passing through the decoder, is output by the fully connected layer to produce the final sequence $Y^{\text{out}} = (y_1^{\text{out}}, y_2^{\text{out}}, y_3^{\text{out}})$.

Experiments

In this section, experiments were conducted on data from four real oil wells to demonstrate the superiority of our proposed method compared with several advanced models. An in-depth analysis also verified the effectiveness of our method.

Data Set Description and Preprocessing. The data used in this paper are from four oil wells. Wells 7P34-P2 and 7P34-P6 are from the Daqing Oil Field in China, Well F14 is from the Volve Oil Field in Norway, and the Z1 data set is from a GitHub project titled "Oil production forecasting using LSTM, gated recurrent unit, and CONV-LSTM seq2vec seq2seq models." The Z1 data set contains only daily oil production data (PRO) and has been preprocessed by the project author. The models and experiments in this study were implemented using MATLAB, a high-level programming language and environment specifically designed for numerical computation, data analysis, and algorithm development.

For Wells 7P34-P2, 7P34-P6, and F14, the data preprocessing involves handling outliers. Each data set includes two columns: time and PRO. The time column represents the daily production hours. A sliding window of 1 year (365 days) with a step size of one quarter (90 days) is applied. Within each window, outliers are identified using the three-sigma method (also known as the rule of three), a widely recognized statistical technique. This method considers data points that exceed three times the median or fall below one-third of the median within each window as outliers. Outliers above this threshold are replaced with the maximum value in the window, while those below are replaced with the minimum value.

The three-sigma method is particularly suitable for this analysis as it effectively handles the variability in production data, ensuring that extreme values do not distort the overall trend. By applying this method, the preprocessing step smooths short-term fluctuations, captures annual trends, and maintains data stability within the quarterly range, thereby improving the accuracy and reliability of subsequent model analyses. Although the Z1 data have been preprocessed by the project author with unknown specific steps, it is still chosen to ensure data diversity and increase the breadth of the study. Using preprocessed data reduces the workload in the data preparation stage and allows for effective comparison and validation with other data.

Next, the feature processing method proposed in this paper is applied to each oilwell data set, adding two new features for each well:

First-order difference (DI) column: First-order difference column is the PRO value of the current row minus the PRO value of the previous row. Since there is no previous value for the first row, the first row of DI uses the value of the second row minus the first row.

First-order sum (S) column: The PRO value of the current row plus the PRO value of the previous row. Since there is no previous value for the first row, the first row of S uses the sum of the first and second rows.

The final training data sets for Wells 7P34-P2, 7P34-P6, and F14 include four columns: time, DI, S, and PRO. For Well Z1, the data set includes DI, S, and PRO. The first-order difference helps remove trends and seasonal components in the time series, making the data more stationary and suitable for various prediction models. The first-order sum feature reflects the cumulative effect of production data, helping the model capture long-term production trends. By introducing DI and S features, short-term fluctuations and long-term trends in oilwell production data can be better captured, improving the model's sensitivity and prediction accuracy. Additionally, smoothing the data within the sliding window reduces noise interference during model training, enhancing the model's stability and predictive performance. These preprocessing and feature extraction methods ensure data quality and consistency, providing a solid foundation for subsequent analysis and modeling.

The final description of the oilwell data used to build the model is shown in **Table 1**. This table includes key statistical measures such as maximum, minimum, median, mean, interquartile range, standard deviation (Std_dev), and range. The range is defined as the difference between the maximum and minimum values, representing the overall spread of the data. The interquartile range represents the range between the first quartile (Q1) and the third quartile (Q3), highlighting the spread of the middle 50% of the data. These measures provide a comprehensive understanding of the variability and distribution of production data across different oil wells.

Well-ID	Data Size	Variables	Maximum	Minimum	Median	Mean	Interquartile Range	Std_dev	Range
7P34-P6	2,472	4	10.180	0.593	3.245	3.397	2.450	1.907	9.586
7P34-P2	2,560	4	10.290	0.003	1.345	1.866	2.740	1.983	10.286
F14	2,723	4	5,644.370	4.300	1,041.770	1,447.753	1,916.430	1,289.762	5,640.070
Z1	5,000	3	2,599.690	670.612	1,621.550	1,660.668	743.486	445.737	1,929.078

DIS, refers to the use of the proposed feature extraction method in this study, including both the differencing feature and the cumulative sum feature. The DI column is calculated as the current row's PRO value minus the previous row's PRO value. For the first row, the DI value is computed as the second row's PRO value minus the first row's PRO value. The S column is calculated as the current row's PRO value plus the previous row's PRO value. For the first row, the S value is calculated as the sum of the first and second rows' PRO values; RAW, Refers to the absence of the proposed feature extraction method, meaning no differencing or cumulative sum features were applied.

Table 1—Basic statistical properties of oilwell data sets.

When conducting the experiments, the last 500 days of data from each oil well were used as the test set, while the remaining data were used for training. After data splitting, it is necessary to perform min-max normalization to prevent the experimental results from being affected by any large differences in the data. The data normalization calculation is shown in Eq. 22:

$$x_i^{stdz} = \frac{X_i - \mu_X}{\sigma_X}, \quad (22)$$

where x_i^{stdz} is the standardized value of the i th feature, X_i is the i th feature value, μ_X is the mean value of that feature, and σ_X is the Std_dev of that feature. After data normalization, each feature value is converted to a standard score, which helps machine learning models better handle features of different scales. Finally, during the prediction phase, the data need to be denormalized back to their original values.

Evaluation Metrics. Common evaluation metrics in time series forecasting include the coefficient of determination (R^2), mean square error, root mean square error, and mean absolute percentage error (MAPE).

This study focuses on multistep forecasting of oil production. Oil production data have different scales and ranges, and the mean and variance of the data may change over the long time series. Additionally, for multistep time series forecasting, the prediction accuracy may vary at different timesteps. Therefore, this paper uses normalized root mean square error (NRMSE), MAPE, and R^2 to evaluate model performance. The definitions of NRMSE, MAPE, and R^2 are shown in Eqs. 23 through 25:

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}}{y_{\max} - y_{\min}}, \quad (23)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%, \quad (24)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (25)$$

where y_i is the actual value of the i th data point in the sequence, $\hat{y}_i$ is the predicted value of the i th data point in the sequence, $\bar{y}$ is the average of the actual values, $y_{\max}$ and $y_{\min}$ represent the maximum and minimum values in the actual sequence, respectively, and N represents the number of samples.

Experimental Results and Analysis

Previous studies have predominantly focused on single-step forecasting methods. In contrast, this research emphasizes multistep forecasting. The following sections present the experimental results and analysis, evaluating the proposed MA-CLSTM-ED model under various settings and comparing it with baseline models.

		7P34-P6		7P34-P2		F14		Z1	
Methods		Training	Test	Training	Test	Training	Test	Training	Test
CNN	R^2	0.8838	0.8675	0.9002	0.8803	0.8244	0.8054	0.9827	0.9776
	MAPE	3.68%	3.92%	8.10%	7.42%	15.01%	25.97%	1.27%	1.49%
	NRMSE	0.0619	0.0626	0.0824	0.0906	0.1116	0.1578	0.0371	0.0436
LSTM	R^2	0.8642	0.8527	0.8956	0.8769	0.7973	0.7549	0.9726	0.9670
	MAPE	4.53%	4.15%	6.77%	7.72%	22.72%	32.93%	1.75%	1.96%
	NRMSE	0.0648	0.0658	0.0843	0.0923	0.1617	0.1958	0.0474	0.0530
CNN-LSTM	R^2	0.8861	0.8666	0.9071	0.8864	0.78695	0.7199	0.9814	0.9739
	MAPE	3.13%	3.50%	8.76%	6.54%	24.28%	33.78%	1.36%	1.61%
	NRMSE	0.0612	0.0627	0.0817	0.0883	0.1717	0.2083	0.0388	0.0472
LSTM-attention	R^2	0.8713	0.8536	0.8992	0.8796	0.7673	0.6931	0.9805	0.9727
	MAPE	4.13%	4.52%	5.57%	7.65%	26.44%	36.96%	1.41%	1.79%
	NRMSE	0.0650	0.0657	0.0812	0.0913	0.1955	0.2174	0.0397	0.0481
Bidirectional LSTM	R^2	0.8694	0.8558	0.8946	0.8744	0.8111	0.7871	0.9790	0.9616
	MAPE	4.10%	4.44%	7.12%	8.02%	20.92%	27.39%	1.44%	2.08%
	NRMSE	0.0645	0.0651	0.0848	0.0930	0.1489	0.1649	0.0413	0.0564
MA-CLSTM-ED	R^2	0.8925	0.8765	0.9101	0.8914	0.8878	0.8632	0.9959	0.9897
	MAPE	3.30%	3.83%	5.53%	7.03%	7.46%	10.60%	1.19%	0.98%
	NRMSE	0.0564	0.0605	0.0801	0.0874	0.0735	0.0781	0.0139	0.0289

Table 2—Comparison of each model when the output step size is 3. Bold values in the table represent the best-performing method for each metric. For example, MA-CLSTM-ED is compared with other methods including CNN, LSTM, CNN-LSTM, LSTM-attention, and Bidirectional LSTM.

Model Performance Comparison. In this section, the performance of the MA-CLSTM-ED model under default parameter settings is assessed and compared with several baseline models. The default parameters are as follows: initial learning rate of 0.01, maximum iterations of 100, batch size of 32, convolution kernel size of 32, CNN filters set to 32, LSTM neurons set to 50, attention heads set to 8, and prediction steps set to 3 (i.e., predicting the values for the next three time points). Under this setting, the MA-CLSTM-ED model proposed in this paper is compared with CNN, LSTM, CNN-LSTM, LSTM-attention, and bidirectional LSTM.

To prevent overfitting and ensure the model's generalization, regularization techniques such as L2 regularization and dropout were used, reducing model complexity by penalizing large coefficients. Additionally, data augmentation was applied to balance the data set by oversampling low-production phases and undersampling high-production phases, thereby enhancing model robustness. A fivefold sliding

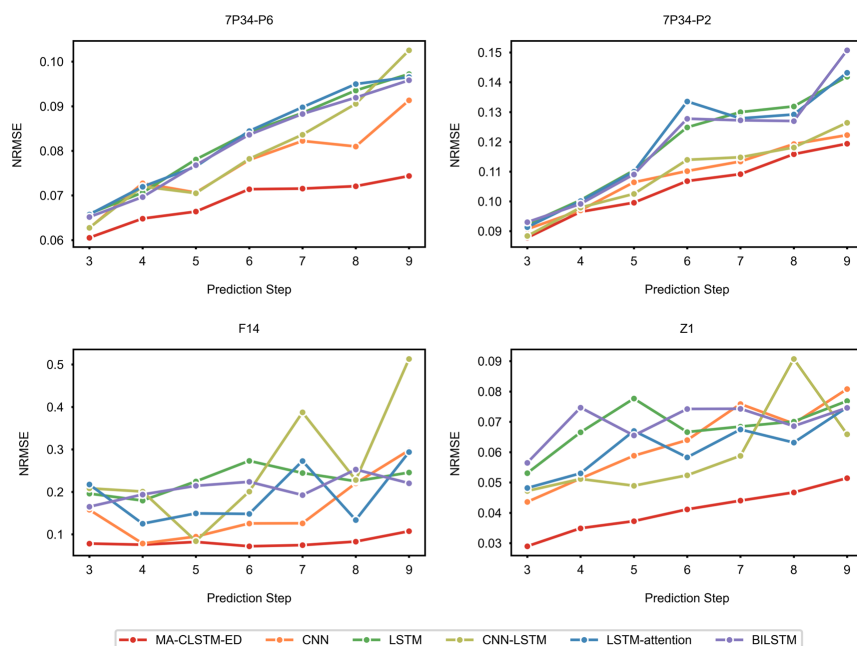


Fig. 5—NRMSE distribution.

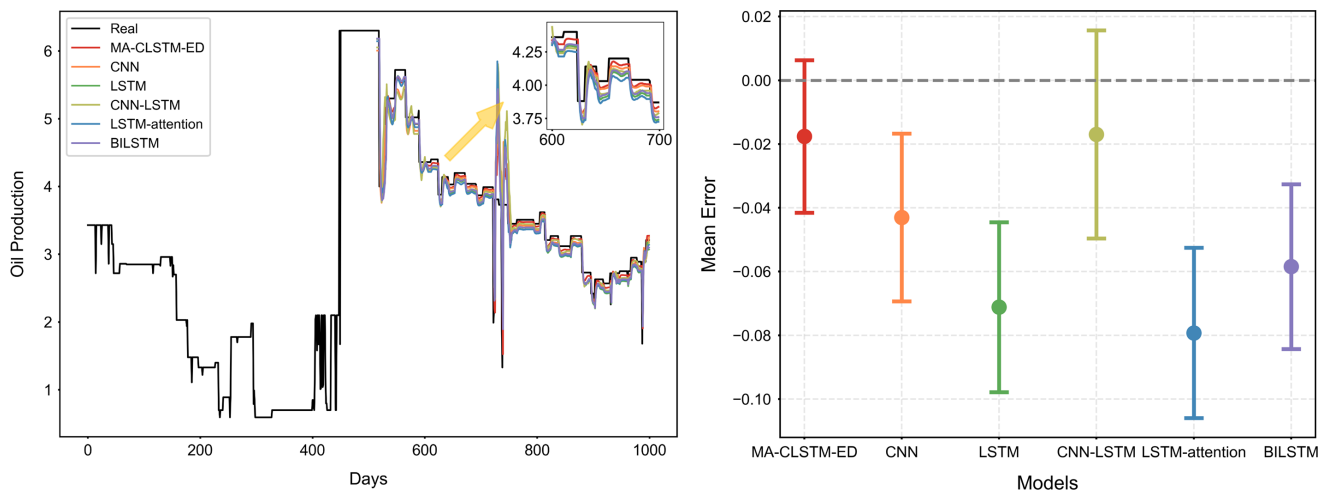


Fig. 6—Comparison results of Well 7P34-P6.

window cross-validation was further implemented to comprehensively evaluate the model's performance across different data segments. The averaged results from these cross-validation folds, showing the final evaluation metrics, are presented in **Table 2**.

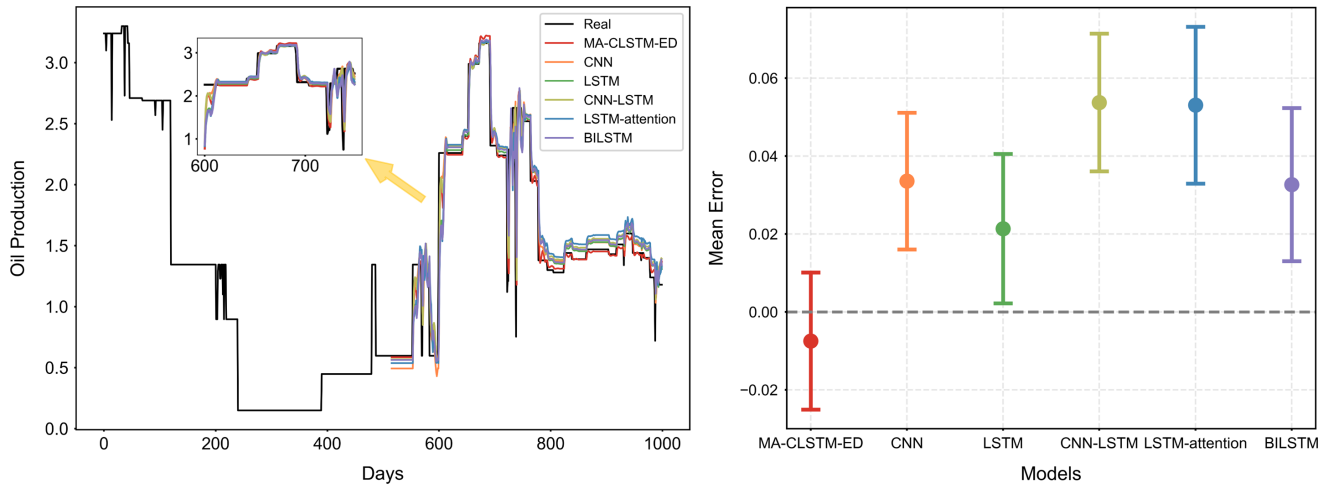


Fig. 7—Comparison results of Well 7P34-P2.

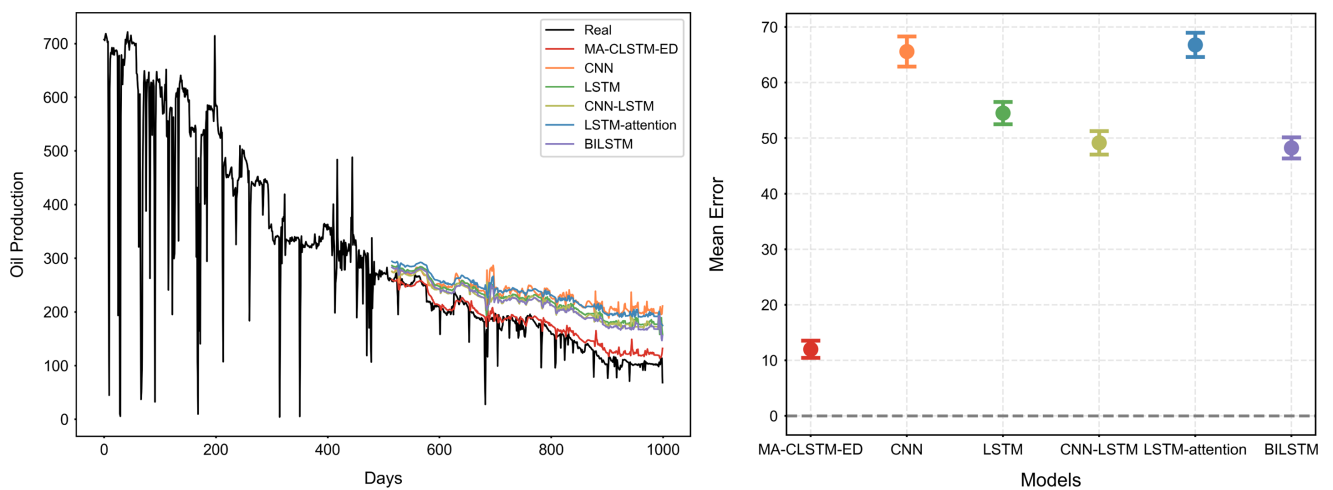


Fig. 8—Comparison results of Well F14.

Subsequently, the output steps were extended, and the specific distribution of NRMSE for each model with output steps of 3, 4, 5, 6, 7, 8, and 9 under default parameter settings was tested. **Fig. 5** illustrates the NRMSE distribution of the proposed model alongside several other comparison models at different output timesteps.

An in-depth comparison of the prediction trends across various oil wells at a prediction step of 9 was conducted to thoroughly evaluate the performance of the proposed MA-CLSTM-ED model, as depicted in **Figs. 6 through 9**. The left-hand side of these figures presents the comparison between the actual oil production data and the predictions generated by different models, including MA-CLSTM-ED, CNN, LSTM, CNN-LSTM, LSTM-attention, and bidirectional LSTM. In addition to this comparison, the analysis was expanded to include the mean prediction errors along with their associated 95% confidence intervals for each model, providing a more comprehensive evaluation. The results, displayed on the right-hand side of **Figs. 6 through 9**, highlight the mean prediction errors with central markers, while the error bars represent the range of the confidence intervals.

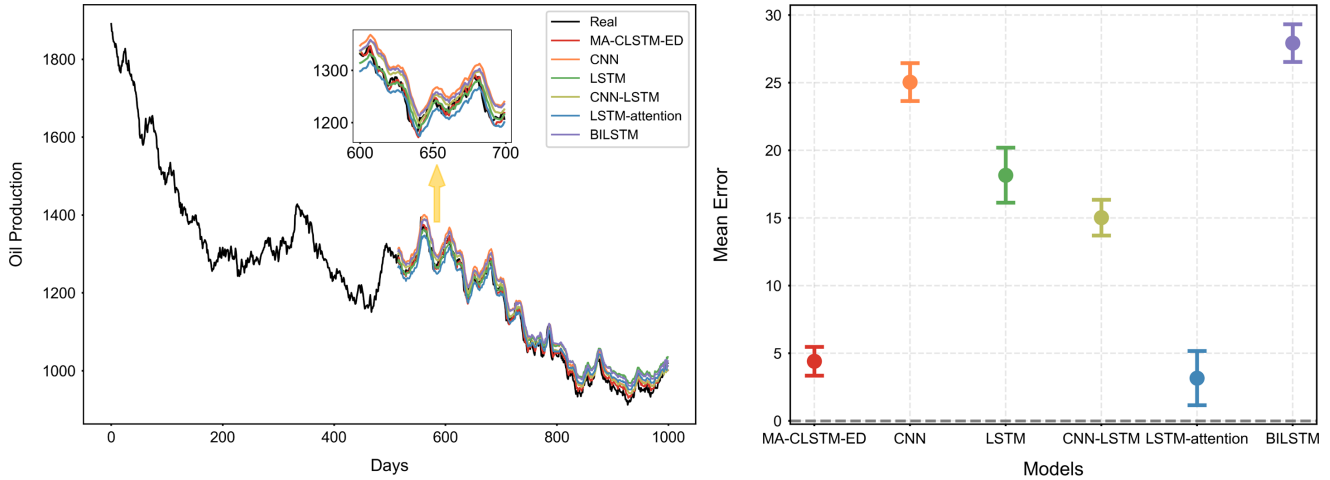


Fig. 9—Comparison results of Well Z1.

To quantitatively assess the confidence levels associated with each model across different wells, the prediction credibility was calculated using a defined equation, as presented in Eq. 26.

$$C_{mi} = \left(1 - \frac{(\hat{y}_{mi} - y_i)^2}{\sum_{n=1}^M (\hat{y}_{ni} - y_i)^2} \right), \quad (26)$$

where C_{mi} represents the i th prediction credibility for the sample by the m th model, $\hat{y}_{mi}$ is the predicted value by the m th model, y_i is the true value, and M denotes the total number of models. Eq. 26 is adapted from the work of Jian et al. (2020), where it was initially introduced to integrate predictions from various machine learning models by assigning weights based on their accuracy in specific data segments. The underlying logic is that models with lower prediction errors relative to others in the ensemble are given higher credibility scores, thus influencing the final prediction more significantly. By using this approach, we ensure that our assessment of model confidence is both rigorous and reflective of each model's comparative performance. The outcomes of this analysis, including the mean confidence values and their Std_dev, are systematically summarized in **Table 3**.

	7P34-P6		7P34-P2		F14		Z1	
Methods	Mean	Std_dev	Mean	Std_dev	Mean	Std_dev	Mean	Std_dev
CNN	0.8504	0.1037	0.8265	0.1644	0.7539	0.0708	0.7693	0.1494
LSTM	0.7820	0.0952	0.8890	0.0779	0.8216	0.0305	0.7953	0.1626
CNN-LSTM	0.8306	0.1528	0.8198	0.0944	0.8635	0.0340	0.9018	0.0779
LSTM-attention	0.7563	0.1056	0.7203	0.1323	0.7194	0.0546	0.8325	0.1772
Bidirectional LSTM	0.8436	0.059411	0.8446	0.0646	0.8629	0.0269	0.7489	0.1052
MA-CLSTM-ED	0.9368	0.0759	0.8995	0.1838	0.9785	0.0669	0.9519	0.0639

Table 3—Comparison of the confidence of each model when the prediction step size is 9.

As shown in **Table 2**, the MA-CLSTM-ED model outperforms the other models across all wells in terms of R^2 , MAPE, and NRMSE metrics, demonstrating its strong ability to handle complex production data variations and capture long-term trends. Notably, the model achieved the highest R^2 and the lowest MAPE and NRMSE values, particularly in Well Z1, where the R^2 reached 0.9959, the MAPE was just 0.98%, and the NRMSE was 0.0289. These results underscore the model's superior predictive accuracy and its effectiveness in minimizing errors and capturing subtle changes in the data. The NRMSE distribution in **Fig. 5** shows that while prediction errors increase

with longer output steps, the MA-CLSTM-ED model maintains a lower error rate compared with others, highlighting its robustness and effectiveness in handling multistep forecasting challenges.

The results from **Figs. 6 through 9** demonstrate that the MA-CLSTM-ED model's prediction trends across different oil wells closely align with the actual data, yielding the lowest mean prediction errors among the compared models. The relatively narrow confidence intervals associated with the MA-CLSTM-ED model indicate greater stability and reliability in its predictions. In contrast, other models exhibit more significant fluctuations in their predictions and broader confidence intervals, suggesting higher uncertainty in their outputs. **Table 3** further illustrates that the MA-CLSTM-ED model shows superior confidence levels at a prediction step size of 9, with the highest mean confidence and the lowest Std_dev across the tested wells. Notably, in the Z1 well, the model achieved a confidence level of 0.9519 with a Std_dev of just 0.0639, highlighting its strong robustness and consistency while maintaining high predictive accuracy.

In summary, the NRMSE of the proposed model is lower than that of other comparison models at most output timesteps, indicating that the MA-CLSTM-ED model has higher prediction accuracy than other models, especially in cases with longer prediction steps. This suggests that the proposed model can effectively capture the complex temporal dependencies in oilwell production data, providing an effective method for predicting oilwell production data.

Ablation Study. To evaluate the impact of the proposed feature extraction method, we conducted a comparative analysis of the NRMSE values for the MA-CLSTM-ED model, both with and without the feature extraction method, under default parameter settings across various output timesteps (3, 4, 5, 6, 7, 8, 9), as presented in **Table 4**.

From **Table 4**, it can be seen that there are certain differences in the NRMSE values for different oil wells under the same prediction step. This may be due to different temporal dependencies and volatility in the production data of different oil wells, affecting the model's performance in handling these differences. The NRMSE values for Oil Wells 7P34-P6 and Z1 are relatively low at all prediction steps, indicating that the data of Oil Wells 7P34-P6 and Z1 have relatively stable temporal characteristics. Moreover, as the prediction step increases, the NRMSE value gradually increases. This means that the longer the prediction step, the greater the uncertainty of future data and the more challenging it is for the model to predict. However, the MA-CLSTM-ED model with the feature extraction method proposed in this paper has lower NRMSE values at different timesteps compared with the model without the feature extraction method, indicating that the feature extraction method has a certain effect on improving the prediction accuracy of the model.

Output Step Size	7P34-P6		7P34-P2		F14		Z1	
	DIS	RAW	DIS	RAW	DIS	RAW	DIS	RAW
3	0.0605	0.0633	0.0878	0.0920	0.0781	0.0798	0.0289	0.0368
4	0.0648	0.0690	0.0964	0.0977	0.0754	0.0972	0.0349	0.0410
5	0.0663	0.0724	0.0995	0.1057	0.0819	0.1036	0.0372	0.0423
6	0.0713	0.0799	0.1068	0.1068	0.0718	0.1396	0.0411	0.0463
7	0.0715	0.0820	0.1091	0.1123	0.0744	0.0902	0.0440	0.0554
8	0.0720	0.0840	0.1158	0.1149	0.0828	0.1037	0.0467	0.0511
9	0.0743	0.0882	0.1193	0.1216	0.1072	0.1415	0.0514	0.0598

DIS, refers to the use of the proposed feature extraction method in this study, including both the differencing feature and the cumulative sum feature. The DI column is calculated as the current row's PRO value minus the previous row's PRO value. For the first row, the DI value is computed as the second row's PRO value minus the first row's PRO value. The S column is calculated as the current row's PRO value plus the previous row's PRO value. For the first row, the S value is calculated as the sum of the first and second rows' PRO values; RAW, refers to the absence of the proposed feature extraction method, meaning no differencing or cumulative sum features were applied.

Table 4—Whether to model loss when applying feature extraction method.

Hyperparameter		Learning_rate	MaxEpochs	MiniBatchSize	KerlSize	FilterSize	NumNeurons	NumHeads
Initial_value		0.01	50	32	32	32	50	8
Initial_range		[0.01, 0.1]	[10, 200]	[16, 512]	[1, 500]	[16, 512]	[1, 200]	[1, 20]
7P34-P6	Final_range	[0.02, 0.04]	[25, 110]	[32, 64]	[161, 201]	[160, 192]	[40, 60]	[9, 11]
	Final_value	0.038	76	56	175	175	57	11
7P34-P2	Final_range	[0.06, 0.08]	[115, 125]	[272, 320]	[341, 381]	[48, 64]	[40, 60]	[4, 10]
	Final_value	0.052	120	304	361	49	59	5
F14	Final_range	[0.04, 0.06]	[90, 130]	[64, 96]	[10, 41]	[80, 128]	[80, 100]	[16, 20]
	Final_value	0.048	98	80	21	112	96	19
Z1	Final_range	[0.02, 0.03]	[60, 100]	[16, 64]	[421, 481]	[278, 320]	[140, 170]	[7, 12]
	Final_value	0.026	80	48	461	304	160	11

Table 5—Value and range of the main hyperparameters of MA-CLSTM-ED model for RIME.

Model Optimization. This section explores the enhancement of the MA-CLSTM-ED model's performance, particularly for long prediction steps, through the application of the RIME (Su et al. 2023) optimization algorithm. By optimizing for a prediction step of 9, the model's prediction accuracy and stability for shorter prediction steps can be indirectly improved. In this process, specific adjustments were made to the parameters for different oil wells. RIME is a novel swarm intelligence-based optimization algorithm that combines the characteristics of soft frost and hard frost. It achieves a balance between global and local searches by simulating the formation and evolution process of frost in nature, with the advantages of strong global search ability, refined local search, and high adaptability. The core mechanism of the RIME algorithm includes the following three parts:

Soft frost search strategy: This is a step-by-step search and usage method that explores the solution space widely by gradually changing the positions of frost agents. This strategy helps discover potential high-quality solutions and avoids local optima.

Hard frost puncture mechanism: This mechanism achieves efficient information exchange between individuals through dimensional crossover and exchange between agents, improving the convergence speed and accuracy of the solution.

Forward greedy selection mechanism: Based on the greedy selection method, the improved forward greedy selection mechanism can increase the diversity of the population and avoid local optima as much as possible.

To simplify, **Fig. 10** shows the basic structure of RIME. The pseudocode for RIME algorithm hyperparameter tuning can be described as follows: (i) initialize the frost population, iterations, and parameter range to be tuned; (ii) obtain the current best agent and best fitness value; (iii) update the population using the step-by-step soft frost search strategy; (iv) implement efficient information exchange between individuals using the hard frost puncture mechanism; (v) retain excellent solutions and eliminate poor ones using the aggressive greedy selection mechanism; (vi) check whether the termination condition is met; if so, end; otherwise, continue iterating. Using this process, the optimal hyperparameters were obtained.

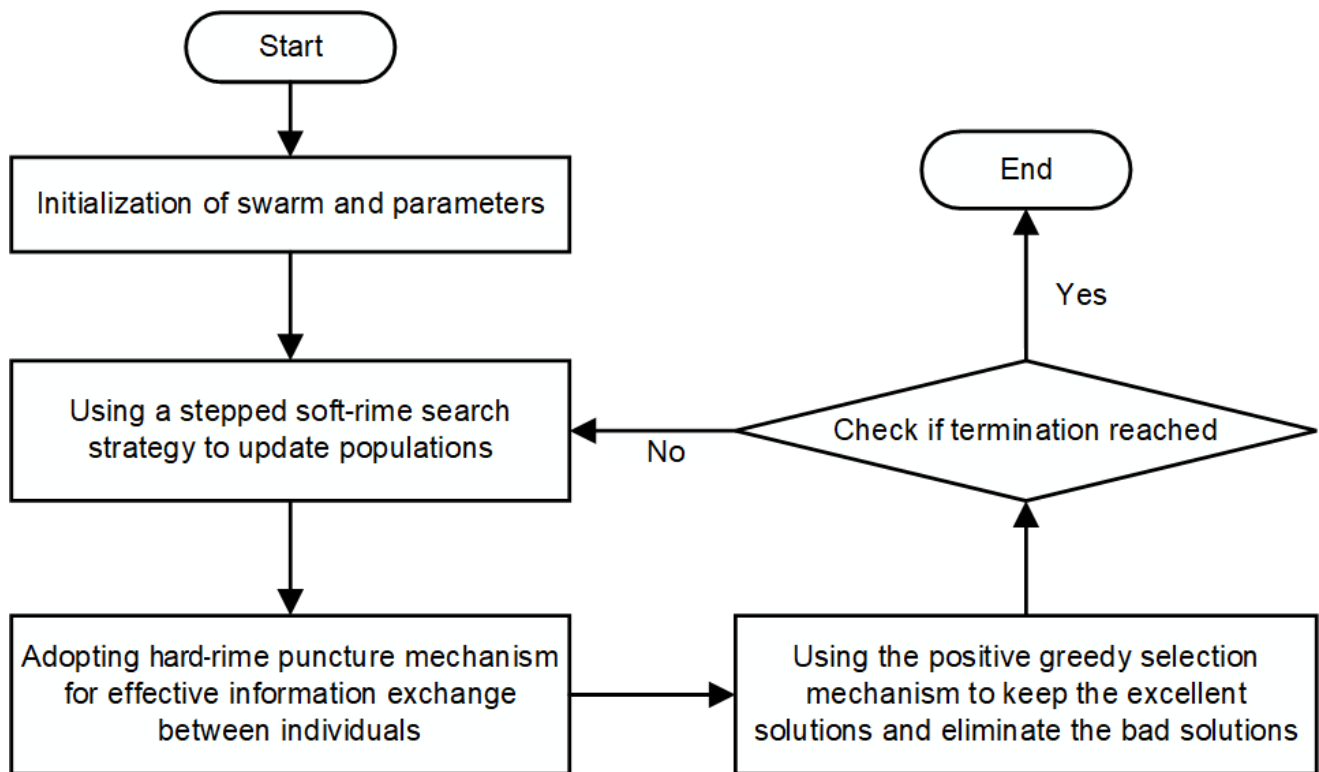


Fig. 10—Flow chart of RIME.

Table 5 shows the initial values and ranges of the default parameters, and the final ranges and parameter values of different oil wells after optimization using the RIME for a prediction step of 9. By adjusting these parameters, we can further enhance the model's performance, thereby verifying the superiority of the proposed model for long prediction steps.

Table 6 shows the performance of the model before and after optimization for different oil wells. From the table, it can be seen that after optimization, the MAPE and NRMSE values for all oil wells decreased, and the R^2 values increased, indicating an improvement in the model's prediction accuracy. By optimizing the MA-CLSTM-ED model for a prediction step of 9, the prediction accuracy of the model for different oil wells was significantly improved, verifying the effectiveness of the feature extraction method and optimization strategy proposed in this paper.

Well-ID	7P34-P6			7P34-P2			F14			Z1		
	R^2	MAPE	NRMSE	R^2	MAPE	NRMSE	R^2	MAPE	NRMSE	R^2	MAPE	NRMSE
Before optimization	0.8285	5.63%	0.0743	0.7678	12.35%	0.1193	0.7168	16.53%	0.1072	0.9661	1.82%	0.0514
Optimized	0.8541	5.24%	0.0687	0.8012	11.32%	0.1160	0.74264	15.91%	0.1039	0.9689	1.68%	0.0488

Table 6—Performance comparison before and after optimization.

Concluding Remarks

In this study, a hybrid deep learning model (MA-CLSTM-ED) combining CNN, LSTM, and multihead attention mechanisms was proposed to improve oil production forecasting accuracy. First-order difference and cumulative sum data were introduced as features, and the RIME was used for hyperparameter tuning.

Experimental results demonstrate that the MA-CLSTM-ED model significantly outperforms other models in oil production forecasting, particularly in multistep forecasting tasks. The model effectively captures short-term fluctuations and long-term trends, as evidenced by superior NRMSE and MAPE indicators. Comparative analysis highlights the crucial role of first-order difference and cumulative sum data in enhancing prediction accuracy. Despite increasing prediction errors with longer prediction steps, the RIME-optimized model maintains high accuracy, validating the effectiveness of the proposed methods.

Despite these promising results, this study has certain limitations that warrant further investigation. First, our model does not incorporate external covariates, such as geological information, operational changes, or environmental factors, focusing solely on the temporal dependencies within production data. While this approach simplifies the model and ensures broad applicability, including such covariates could potentially enhance the model's predictive accuracy and robustness across diverse production environments. Second, the current model provides deterministic forecasts, lacking the capability to generate probabilistic predictions. Probabilistic forecasting is crucial for quantifying the inherent uncertainty in oil production, thereby supporting more informed decision-making. Lastly, while the MA-CLSTM-ED model is designed to be broadly applicable across different wells and production scenarios, there is a potential risk that it may not generalize well to new or unseen operational conditions that differ significantly from the data used for training. This could limit the model's effectiveness in highly variable environments or in cases where external factors play a significant role. To address these limitations, future research will explore the following directions:

Expanding feature use: While this study focused on the use of first-order difference and cumulative sum features derived from production data, future work could explore applying these features to other relevant variables, such as geological and operational data, to assess their impact on forecasting accuracy. By incorporating a broader set of features, we aim to capture more complex relationships and improve the model's overall predictive performance.

Developing probabilistic forecasting capabilities: Future efforts will focus on extending the model to provide probabilistic forecasts, which can better account for the uncertainties in oil production and contribute to more robust risk assessment and strategic planning. Implementing probabilistic methods will enhance the model's ability to support decision-making processes under uncertainty.

Enhancing generalization through transfer learning: To improve the model's ability to generalize across different operational conditions and unseen scenarios, future research could explore the application of transfer learning techniques. By leveraging pretrained models and fine-tuning them on new data sets with varying production characteristics, we aim to enhance the model's adaptability to diverse and evolving production environments.

This study can serve as a submodule of an intelligent oilfield management system, providing oil production forecasts and playing an important role in production decision-making. Despite the successes achieved in this study, future work should consider integrating more geological information, downhole data, and environmental factors to further improve the model's prediction performance and practical utility.

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